

Barrett Larson

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EDUCATION

Virginia Tech, College of Engineering

Blacksburg, VA

Bachelor of Science in Computer Science, Minor in Mathematics | GPA: 3.70

Expected May 2027

- **Relevant Coursework:** Data Structures & Algorithms (Java), GUI Programming (JavaScript), Database Management Systems (SQL), Computer Systems (C)

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Java, SQL, C, HTML, CSS

Frameworks: React.js, Node.js, Express.js, FastAPI, Flask, Spring Boot

Databases & Cloud: MySQL, MongoDB, AWS (Lambda, SNS, CloudFormation), AWS Cloud Practioner

EXPERIENCE

Expedition Technology

Herndon, VA

Software Engineering Intern

May 2026 – Present

- Implementing frontend and backend features for a storefront platform for buying and selling AI training data.
- Developing a collaborative annotation editor prototype using WebSockets and FastAPI.

CARFAX

Centreville, VA

Software Developer Intern

May 2025 – August 2025

- Developed and deployed an AWS Lambda function that queried Salesforce to retrieve account and contact data, publishing results to dedicated AWS SNS topics.
- Upgraded backend codebases to Java 21 with Spring Boot 3.5+, replacing deprecated libraries and resolving runtime-breaking changes across multiple microservices to address security vulnerabilities.
- Refactored legacy Node.js Lambda functions to Python, resolving critical security vulnerabilities and improving reliability with pytest-based unit tests.
- Automated infrastructure provisioning using AWS CloudFormation (YAML) and implemented CI/CD pipelines with GitLab CI, reducing manual deployment effort and improving test reliability.

PROJECTS

Fraternity Formals Ticket Platform | *MongoDB, Express.js, React.js, Node.js*

- Built a custom ticket purchasing platform for chapter formals, eliminating third-party ticketing fees and saving over \$1,000 per event.
- Integrated Stripe for secure payment processing with a streamlined checkout flow, guest name collection, and automated email confirmations.
- Designed a minimal-friction user experience from ticket selection to purchase confirmation, reducing drop-off and simplifying event management.

Restaurant Management System | *Python, Flask, MySQL, React.js*

- Developed a full GUI application for managing a restaurant's DBMS, including employees, suppliers, recipes, ingredients, orders, and item inventory.
- Implemented complete CRUD operations across all entities using a RESTful Flask backend with structured MySQL relational schemas and foreign-key constraints.
- Built secure employee and admin authentication with hashed passwords, session persistence, and role-based access control for protected operations.

HOBBIES & INTERESTS

Weightlifting, Piano, Basketball (Intramural), Chess, Disc Golf, Catan, Reading